

Mech support\test2.cpp

```

1  /*****
2
3  Q2. Create a C++ class called Robot to control a robot.
4
5  The robot has the following properties:
6  - Speed
7  - Battery
8
9  Requirements:
10 1. Make the data members private.
11 2. Create the following member functions:
12   - setSpeed()
13   - setBattery()
14   - moveForward()
15   - moveBackward()
16   - turnLeft()
17   - turnRight()
18   - displayStatus()
19 3. Speed should only accept values between 0 and 100.
20 4. Battery should only accept values between 0 and 100.
21 5. The robot should not move if the battery is 0%.
22 6. Each movement should reduce the battery by 5%.
23 7. Create a Robot object in main().
24 8. Take the speed and battery values as input from the user.
25 9. Execute a sequence of movement commands.
26 10. Display the final speed and battery level.
27
28 Example:
29
30 Input:
31 Speed: 80
32 Battery: 100
33
34 Commands:
35 Forward
36 Left
37 Forward
38 Right
39
40 Output:
41 Robot Speed: 80
42 Battery: 80%
43
44 code by Rayan
45
46 *****/
47 #include <iostream>
48 using namespace std;

```

```
49
50 class Robot{
51     private:
52         double speed;
53         double Battery;
54     public:
55         void setSpeed(double vel){
56             if(vel>=0 && vel <=100){
57                 speed = vel;
58             }
59         }
60         void setBattery(double bat){
61             if(bat>=0 && bat <=100){
62                 Battery = bat;
63             }
64         }
65         void moveForward(){
66             if(Battery >0){
67                 Battery-=5;
68                 cout << "Moved Forward\n";
69
70             }else{
71                 cout << "Low Battery\n";
72             }
73         }
74         void moveBackward(){
75             if(Battery >0){
76                 Battery-=5;
77                 cout << "Moved Backward\n";
78
79             }else{
80                 cout << "Low Battery\n";
81             }
82         }
83         void turnRight(){
84             if(Battery >0){
85                 Battery-=5;
86                 cout << "Turned Right\n";
87
88             }else{
89                 cout << "Low Battery\n";
90             }
91         }
92         void turnLeft(){
93             if(Battery >0){
94                 Battery-=5;
95                 cout << "Turned Left\n";
96
97             }else{
98                 cout << "Low Battery\n";
```

```
99         }
100     }
101     void dispSpeed(){
102         cout << "Speed: " << speed<<endl;
103     }
104     void dispBattery(){
105         cout << "Battery: " << Battery<<endl;
106     }
107 };
108
109 double Speed;
110 double batt;
111
112 int main()
113 {
114     Robot r;
115     cout <<"Speed: ";
116     cin >> Speed;
117     cout << "Battery: ";
118     cin >> batt;
119     r.setSpeed(Speed);
120     r.setBattery(batt);
121     r.moveForward();
122     r.turnLeft();
123     r.moveForward();
124     r.turnRight();
125     r.dispSpeed();
126     r.dispBattery();
127     return 0;
128 }
```